Skills Summary

A Fraction of a Whole Number or Set

Use this reference while completing your worksheet or when studying.

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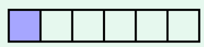
1. One equal group: a unit fraction of a set

The bottom number makes the groups. It tells you how many *equal* groups to break the whole set into, so you **divide** by it.

$\frac{1}{5}$ of 20 means: break 20 into 5 equal groups ($20 \div 5 = 4$) and take one group. The answer is how many things are *in* a group, not how many groups there are.

Example: What is $\frac{1}{6}$ of 30 counters?
Step 1. The top number is 1, so this is just "make the groups and take one" — only the bottom number does any work.
Step 2. Break 30 into 6 equal groups: $30 \div 6 = 5$ in each group.



One group holds **5** counters

Try it: What is $\frac{1}{4}$ of 28 marbles?

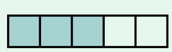
check: **7**

2. Several equal groups: divide, then multiply

Divide by the bottom, then multiply by the top. The top number says how many of the equal groups to take.

$\frac{3}{4}$ of 20: $20 \div 4 = 5$ in each group, then $3 \times 5 = 15$.
Always in that order. Dividing first keeps the numbers small and keeps the picture true to the words.

Example: What is $\frac{3}{5}$ of 40 stamps?
Step 1. The bottom number 5 says five equal groups and the top number 3 says take three of them, so this needs both moves, not just the division.
Step 2. Divide: $40 \div 5 = 8$ stamps in one group. Multiply: 3×8.



Three groups of eight: **24** stamps

Try it: What is $\frac{2}{7}$ of 42 beads?

check: **12**

3. Working backwards to find the whole

When the *part* is known and the whole is missing, run the two moves in reverse:

Divide the part by the top number to get one group, then **multiply by the bottom number** to rebuild the whole.

Forwards you divide then multiply; backwards you divide then multiply too — but by the other numbers. Draw the bar and it is hard to mix up.

Example: $\frac{2}{3}$ of a number is 18. What is the number?
Step 1. The 18 covers two groups, not one, so it cannot be multiplied by 3 straight away — find one group first.
Step 2. One group: $18 \div 2 = 9$. Whole: 3×9, since the bottom number says the whole is three groups.

The number is **27** (check: $27 \div 3 = 9$ and $2 \times 9 = 18$)

Try it: $\frac{3}{4}$ of a number is 21. What is the number?

check: **28**

(!) Watch out: do not divide by the top number by mistake — the *bottom* number always makes the groups. And do not stop after dividing: if the top number is bigger than 1, there is a multiplication still to do.